Sources of $\vec{B}$

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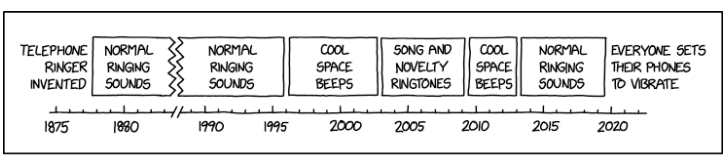

AFTER 140 YEARS, HUMANITY IS FINALLY ON THE VERGE OF WINNING THE WAR AGAINST RINGTONES.

1 Tidbits of History of Magnetism

- Items that measures magnetic field were called loadstones
- Animals used magnetism to orient themselves naturally far before we did
- Funny and 'spooky' idea of affects at a distance – magnetism and love, infection etc. all act at a distance, maybe they're related?
- Newton tinkered with magnets but didn't do anything too unique with them.
- Coulomb actually made some developments on them.
- The French made basically most of the initial developments on magnetism in the Reniassance.

2 Maxwell's Equations

(of magnetism)

Gaussian surfaces:

$$\oiint \vec{E}.d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$\oiint \vec{B}.d\vec{A} = ?$$

Ampere's Loops:

$$\oint \vec{E}.d\vec{S} = -\frac{d\phi}{dt}B$$

$$\oint \vec{B}.d\vec{S} = M_0 I_{thru} + M_0\epsilon_0\frac{d\phi}{dt}\epsilon$$

3 Basic Properties of Magnets

Some small points to begin:

- Instead of plus & minus, we have north & south
- Opposites attract, likes repel
- Appears impossible to get north or south alone. One cannot live while the other survives!

- Some objects (i.e. ferromagnets) are attracted to both ends of a magnet
- Magnets do not attract/repel charges

3.1 Gauss' Law Differences

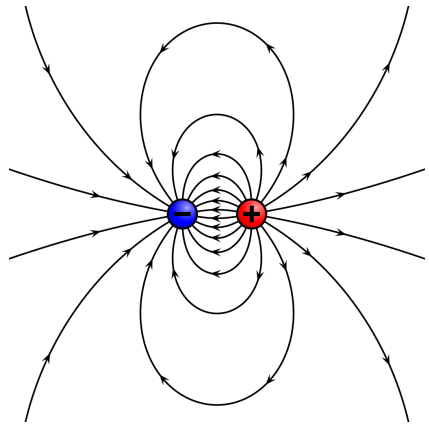

Figure 1: Dipole example

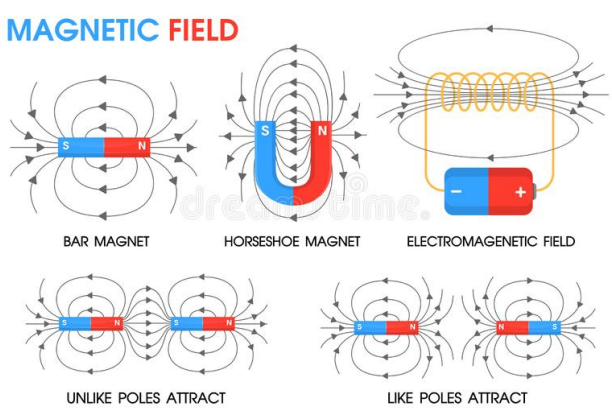

Figure 2: Magnetic Field Flux

In magnetism, the field goes *through* the magnet. It enters the south pole, then travels north. Flux through a closed Gaussian surface is zero.

3.2 Gauss' Law for Magnetism

$$\phi_B = \oiint \vec{B}.d\vec{A} = 0$$

The middle double surface integral represents flux through a closed gaussian surface. Therefore, this intends there is no magnetic equivalent of charge, since magnetic monopoles do not exist in known science.

It is therefore impossible to draw a gaussian surface with net ϕ_B. What causes magnetic fields, then?

3.3 Source of Magnetic Fields

A Danish physicist by the name of Hans Christian Oested proved that currents generate magnetic fields.

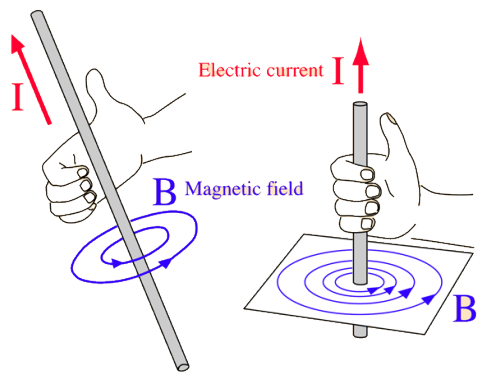

Figure 3: Magnetic field from current

Right-hand Rule #1

1. Use right hand and point thumb in direction of current
2. Curl fingers around wire (not actually)
3. Fingers will point in direction of field

The magnetic field will always be perpendicular-to-the-right of current direction.

4 Equations

Magnetic field equations and their electric field equivalents

4.1 Magnetic field from a point charge

$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r} === \vec{B} = \frac{M_0}{4\pi} q \frac{\vec{v} \times \vec{r}}{r^2}$$

Or, with the cross-product completed:

$$B = \frac{M_0}{4\pi}\frac{qv}{r^2}\sin\theta$$

with theta being the angle between $\vec{v}$ and $\vec{r}$.

4.2 Other formulae

- Magnetic constant
$$M_0 = 4\pi \times 10^{-7}\frac{N}{A^2}$$
- Infinite wire
$$B = \frac{M_0 I}{2\pi r}$$
- Loop of current (center)
$$B_{coil} = \frac{M_0 N I}{2R}$$
with N being the number of turns